

TEST SESSION READINESS

This document provides instructions on configuration of the Secure Exam Browser (SEB).

This is also a place where you could include procedures or checklists related to your internal process for preparing Tests Sessions.

Examples of the types of items you may want to track are available on request from [CAEC Testing Service Support](#).

Testing Computer Requirements

System Details

Operating System (and Devices)	Supported Browsers (listed versions are recommended)	Required Secure Environment
Windows (Microsoft)	Google Chrome – version 62.0 to 115.0 Mozilla Firefox – version 78.0 to 116.0 Microsoft Edge – version 80.0 to 115.0	Safe Exam Browser
Mac OSX 10.13.6 High Sierra or higher (Apple, MacBook)	Safari – version 10.1 to 16.5 Google Chrome – version 62.0 to 115.0 Mozilla Firefox – version 78.0 to 116.0	
Mac iOS iPad Model 7 or higher (Apple, iPad)	Safari – version 10.1 to 16.5	

The minimum recommended screen resolution for the devices to run the assessment is **768 x 1024**. This allows any device the size of an iPad Mini or larger to be used.

The minimum recommended memory for each Operating System is listed below.

System Type	Minimum RAM Required
Windows	6GB
Mac	6GB
iPad	4GB

Bandwidth and Connectivity Requirements

The assessment requires a connection that does not go below 80 kbps per device. For optimal functioning, an average of 2 Mbps or greater is recommended.

IT teams are responsible for determining and reporting the bandwidth and connectivity at the Testing Locations. The following are questions you may want to ask technical staff handling setup at the Testing Locations:

- How are computers connected to the Internet?
 - Are they wired or do they use Wi-Fi?
- What is the bandwidth available to each device that will be used in the assessment?

Firewall's domain-allowed list

The following addresses are used by the assessment site and need to be allowed to run on the assessment devices at all Testing Locations.

Please add these addresses to the firewall's domain-allowed list:

- <https://abed.vretta.com>
- <https://abed-api.vretta.com>
- <https://fonts.googleapis.com>

- <https://s3.ca-central-1.amazonaws.com>
- <https://d3f6c695rnoy7r.cloudfront.net>
- <https://d3azfb2wuqle4e.cloudfront.net>
- <https://cdnjs.cloudflare.com>
- <https://fonts.gstatic.com>
- <https://firebasestorage.googleapis.com>
- <wss://mqvurtmevc.execute-api.ca-central-1.amazonaws.com>
- <https://www.wiris.net>
- <https://svc.webspellchecker.net/spellcheck31/api>

IP Address Considerations

The IP addresses used may change, and firewall rules should be domain-based, not IP-based. If IP-based rules must be used, any IP address within the IP ranges used by AWS should be allowed.

DNS Considerations

The application expects that browsers will resolve domain names using publicly available DNS based on the authoritative name server for a given domain.

Browsers

The technology used to run the test follows modern security and performance standards; however, it is important to ensure that the computers/devices being used for the test are technically suitable for running the test.

The screens that the Invigilators use on test day should be at least 768 x 1024 pixels.

Please consult the chart below to identify which browsers and operating systems are compatible with the test website:

Supported Browsers (listed versions are recommended)
Google Chrome – version 62.0 to 115.0 Mozilla Firefox – version 78.0 to 116.0 Microsoft Edge – version 80.0 to 115.0
Safari – version 10.1 to 16.5 Google Chrome – version 62.0 to 115.0 Mozilla Firefox – version 78.0 to 116.0
Safari – version 10.1 to 16.5

Safe Exam Browser (SEB)

The test can only be accessed via the SEB, using a valid configuration file provided from the test website. The SEB program is a specialized web browser that restricts test applicants' access to tools and resources while they are taking the test. The SEB configuration file regulates access to system functions, other applications, and other settings, while also pointing the SEB program to the test and performing a special "handshake" to validate security.

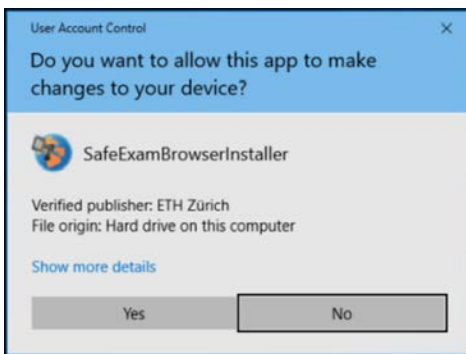
Installing SEB on Windows

1. Download the Safe Exam Browser Installer from the SEB website at https://safeexambrowser.org/download_en.html#WindowsLegacy.

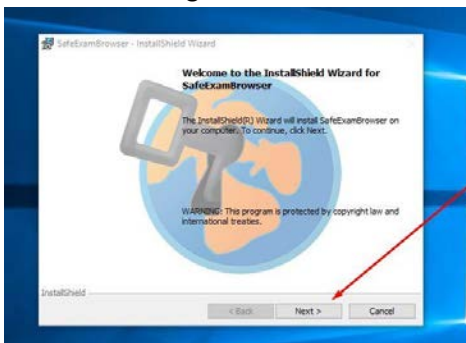
Click the download link for the Safe Exam Browser for Windows.

2. Once the download is complete, run the "SafeExamBrowserInstaller.exe" file to initiate the installation.

You will be asked to allow the installation to make changes on your device. You must accept this to proceed.

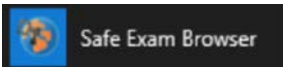


3. You will be presented a Wizard. Click the "Next" button to proceed. Select "I accept the terms in the license agreement", "Install", and "Finish".



You may be prompted to navigate to your installation of Zoom. This is optional and can be skipped if you don't need it.

4. Click "Start" and find "Safe Exam Browser" to run the program.



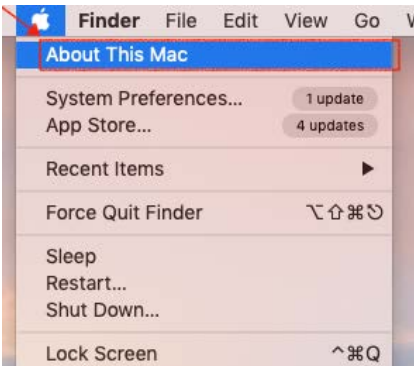
5. You will likely see the picture below with the alert: "You have installed Safe Exam Browser (SEB), but it isn't configured yet." This indicates that the installation of the program is complete.

For instructions on how to access the test, please see the section "Loading the SEB Configuration File (for Computer Labs)".



Installing SEB on macOS

1. Navigate to the "About This Mac" menu in your system menu bar to see if the device is compatible with the SEB.



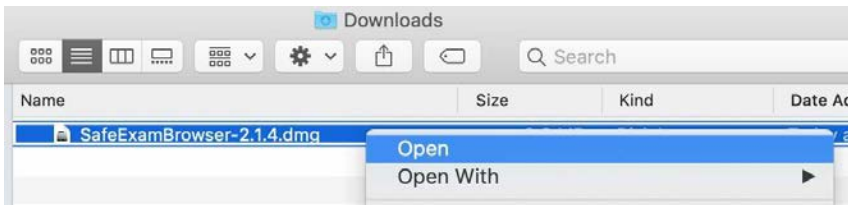
2. You will see a window containing the version of your operating system. It must be higher than Mac OS X 10.13.6 to run the Safe Exam Browser for the exam you are taking. In most cases, it is possible to upgrade the operating system at no cost, but it can take more than an hour.



3. Download the Safe Exam Browser Installer from the SEB website at https://safeexambrowser.org/download_en.html#MacOSX.

Click the download link for the Safe Exam Browser for macOS.

4. Double click on the .DMG file.



5. Drag the Safe Exam Browser icon and drop it in the Applications folder icon to the right. You will be asked for an administrator username and password for the computer.



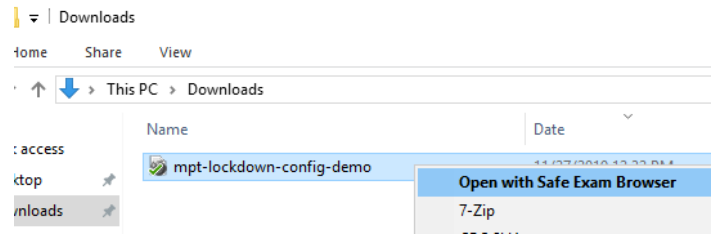
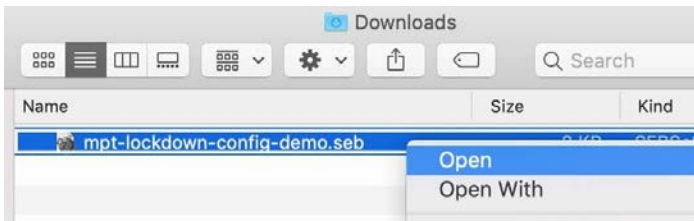
6. You can verify the installation by opening Safe Exam Browser (via Spotlight). The Safe Exam Browser should open at full screen with a message that it needs settings or a link to continue. This indicates that the installation of the program is complete.

For instructions on how to access the test, please see the section "Loading the SEB Configuration File (for Computer Labs)".

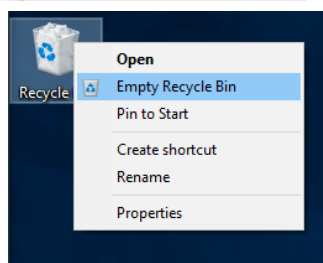
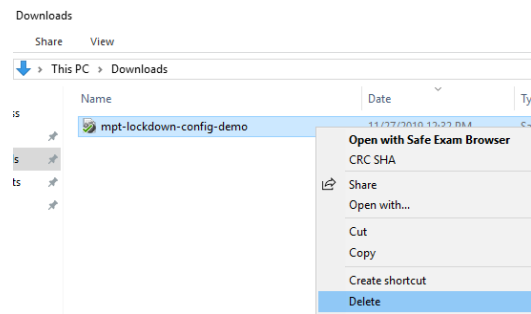
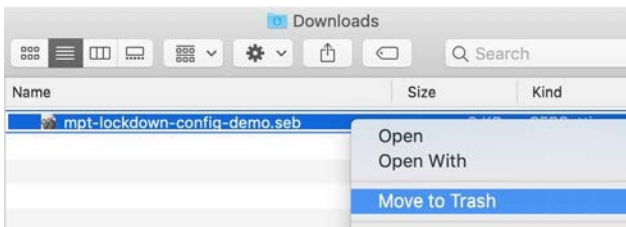
Loading the SEB Configuration File (for Computer Labs)

In order to access the test, the website needs to be opened through the Safe Exam Browser using a test applicant SEB configuration file. This file opens to a login screen which then opens directly to the test.

When the computer lab is being set up for the test, all of the computers should be opened to SEB prior to test applicants entering the room. Please note that the software should not be opened directly, but by double-clicking on the configuration file.



Please Note: The configuration file must be completely removed from each computer before the last test administrator leaves the test room.



Loading the SEB Configuration File (for BYOD)

In order to access the test, the website needs to be opened through the Safe Exam Browser using a test applicant SEB configuration file. This file opens to a login screen which then opens directly to the test.

In a BYOD test administration, test applicants download their version of the SEB configuration file from within the website. The SEB file should never be provided directly to test applicants by the Test Administrator. The configuration file that is provided directly to test applicants has reduced security, which is accommodated for during the test session by preventing their start of the test until their identity and presence have been verified.

Please Note: Chromebooks and iPads are not supported at this time.